



Listening at Scale

Leveraging AI to Analyze Diplomate Feedback on MOCA Minute items

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Disclosure

Nothing to Disclose



Session learning objectives

- **Analyze at scale**
 - How AI can analyze large volumes of diplomate feedback on longitudinal assessments such as MOCA Minute
- **Improve continuously**
 - How AI-enabled analysis supports continuous improvement within a continuing certification system
- **Collaborate across teams**
 - The role of assessment, product management, and AI teams in producing transparent, feedback-informed reporting
- **Earn trust**
 - How acting on diplomate feedback strengthens a board's role as a trusted partner in lifelong learning



The feedback loop built into MOCA Minute

Every year, tens of thousands of board-certified anesthesiologists answer 120 MOCA Minute questions to maintain certification. After each question, they rate their confidence and the relevance of the item to their practice — and can leave open-ended comments.



The comments have always been there. What's changed is our ability to review and analyze all of them.

2025 MOCA Minute Diplomate Feedback Summary Report

MOCA Minute questions distributed in 2025	4,831
Questions with comments	2,954
Number of comments	24,474
Number of diplomates who submitted comments	8,366
Questions revised based on comments	167
Questions deleted based on comments	99

Most Commented Content Categories

ABA Blueprint Code	Level 1	Level 2
II.A.4. Substance Use Disorder	Pharmacology	General Concepts
II.B.1. Central and Peripheral Nervous System	Clinical Sciences: Anesthesia Procedures, Methods and Techniques	Perioperative Management of Patients with Chronic Disease States
II.B.8. Neurovascular Diseases	Clinical Sciences: Anesthesia Procedures, Methods and Techniques	Perioperative Management of Patients with Chronic Disease States
IV.D.3. Pharmacology	Organ-Based Basic and Clinical Sciences	Gastrointestinal/Hepatic Systems
V.C.6. Special Considerations in Obstetric Anesthesia	Clinical Subspecialties	Obstetric Anesthesia
V.H.9. Postoperative Care and Transitions of Care	Clinical Subspecialties	Geriatric Anesthesia/Aging

V.L.4. Immune Dysfunction/Reaction	Clinical Subspecialties	Critical Care Anesthesia
V.Q.2. Pharmacology	Clinical Subspecialties	Regional Anesthesia
V.Q.3. Regional Anesthesia	Clinical Subspecialties	Regional Anesthesia
V.E.9. Diversity, Equity, and Inclusion (DEI) in the Healthcare Workplace	Special Problems or Issues in Anesthesiology	Ethics, Practice Management and Medical/Legal Issues

Themes

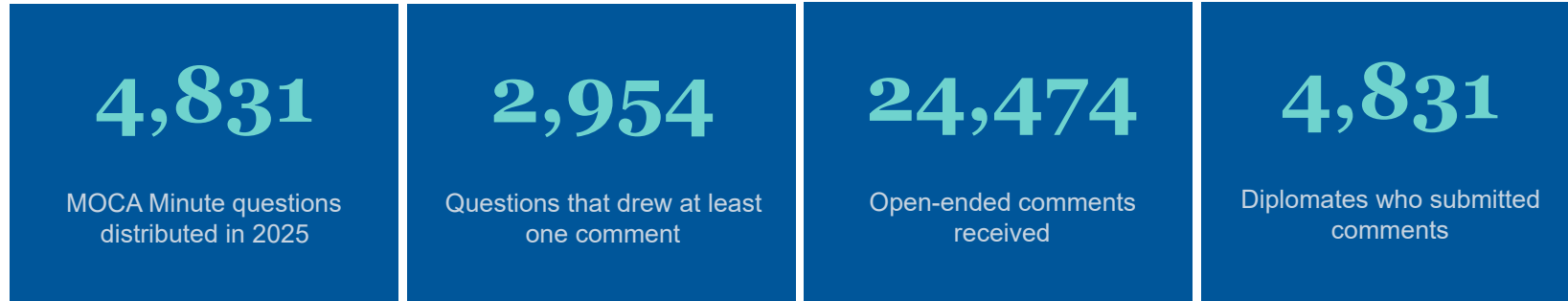
Theme	Description
Inaccurate or Inapplicable Questions	Answers perceived as incorrect, outdated, or misaligned with current clinical guidelines or real-world anesthesiology practice, including misclassified concepts or unrealistic scenarios.
Clarity and Wording Issues	Confusing, ambiguous, or misleading wording especially regarding timing, rationale, or phrasing.
Professional Disagreement or Controversy	Differing opinions on correct answers, clinical best practices, or interpretation of evidence, including conflicting guidelines or lack of consensus.
Social Determinants of Health and DEI Relevance	Criticism about the inclusion, frequency, and relevance of questions on social determinants of health, race, ethnicity, and DEI topics.
Praise and Positive Feedback	Positive sentiments about question quality, educational value, or overall experience.
Clinical Relevance to Practice	Questions viewed as lacking practical value or real-world usefulness, particularly those involving rapidly evolving guidelines or outdated techniques.

Note: Three questions that were distributed as part of an Article-Based Question pilot with ASA were excluded from the report analysis. Comments associated with those items are being evaluated separately.

MOCA Minute Feedback Report

- Report is an initiative in the ABA's 2025-2030 Strategic Plan under Strategy 2: Develop a Seamless Continuing Certification System
- Aggregates diplomate feedback on MOCA Minute items, identifies key themes and outlines actions taken, including question revisions and removals
- Some of the data is available via current reporting capabilities, although identification of themes required the use of AI
 - The ABA receives more than 20,000 comments on MOCA Minute items annually, which is too large a volume for staff to effectively review and analyze on an aggregate basis.

By the numbers



- **Why manual review does not scale**

- Comments are free text, unstructured, and highly variable in length and usefulness.
- Item-level review catches problems one question at a time — it cannot surface patterns across the pool.

Analysis was narrowed to items with 500+ first-time responses, still leaving roughly 14,000 comments.

How the AI analysis works

From raw comments to a short list of validated themes

A repeatable pipeline, not a one-off experiment

Scope

Restrict to items with 500+ first-time responses; exclude the Article-Based Question pilot.

Clean

Discard non-substantive and offensive text ('ok', 'I', 'u') so signal is not diluted.

Surface Themes

Prompt large language models to propose candidate themes, categories, and topics.

Consolidate

Run multiple model architectures and compare results, mimicking majority voting.

Validate

Assessment and product staff review the long and short lists and adjust definitions.

Where AI fell short — and how people fixed it

Diplomate comment:

"How does this question help me take care of my patients?"

What the model got wrong

- The model coded this as a question-clarity issue because the comment referenced “this question”
- It is actually about relevance to practice, not wording
- The model pattern-matches; it does not reason its way to “wait — this is about relevance”
- Praise and positive feedback were initially overlooked entirely

How the teams fixed it

- Rebuilt the architecture to trace each theme back to its originating comments and item IDs.
- Applied natural language processing techniques to better instruct the AI model
- Authored human-defined themes such as Question Value and Relevance to Practice to capture the nuance.
- Constrained long and short lists to reduce redundancy and strengthen prevalent themes.
- Reviewed both lists by hand, promoting important themes the model left behind.

Validating output against human ground truth

Exam editors' own item-level summaries served as ground truth. AI summaries were compared to them both by human review and by automated similarity scoring across ten methods.

Human review

Editors compared AI summaries with their own line by line, marking exact overlap and any added context.

Similarity scoring

Lexical, embedding, and transformer models scored how closely AI summaries matched the ground truth.

Iterative re-runs

AI staff refactored code to reduce runtime by about 80% allowing results to be regenerated and re-checked more rapidly.

Lesson: treat model output as a draft that must earn its place in the report. Powerful general-purpose models took us far; closing the remaining gap will require more in-house natural language processing tuned to our nuances.

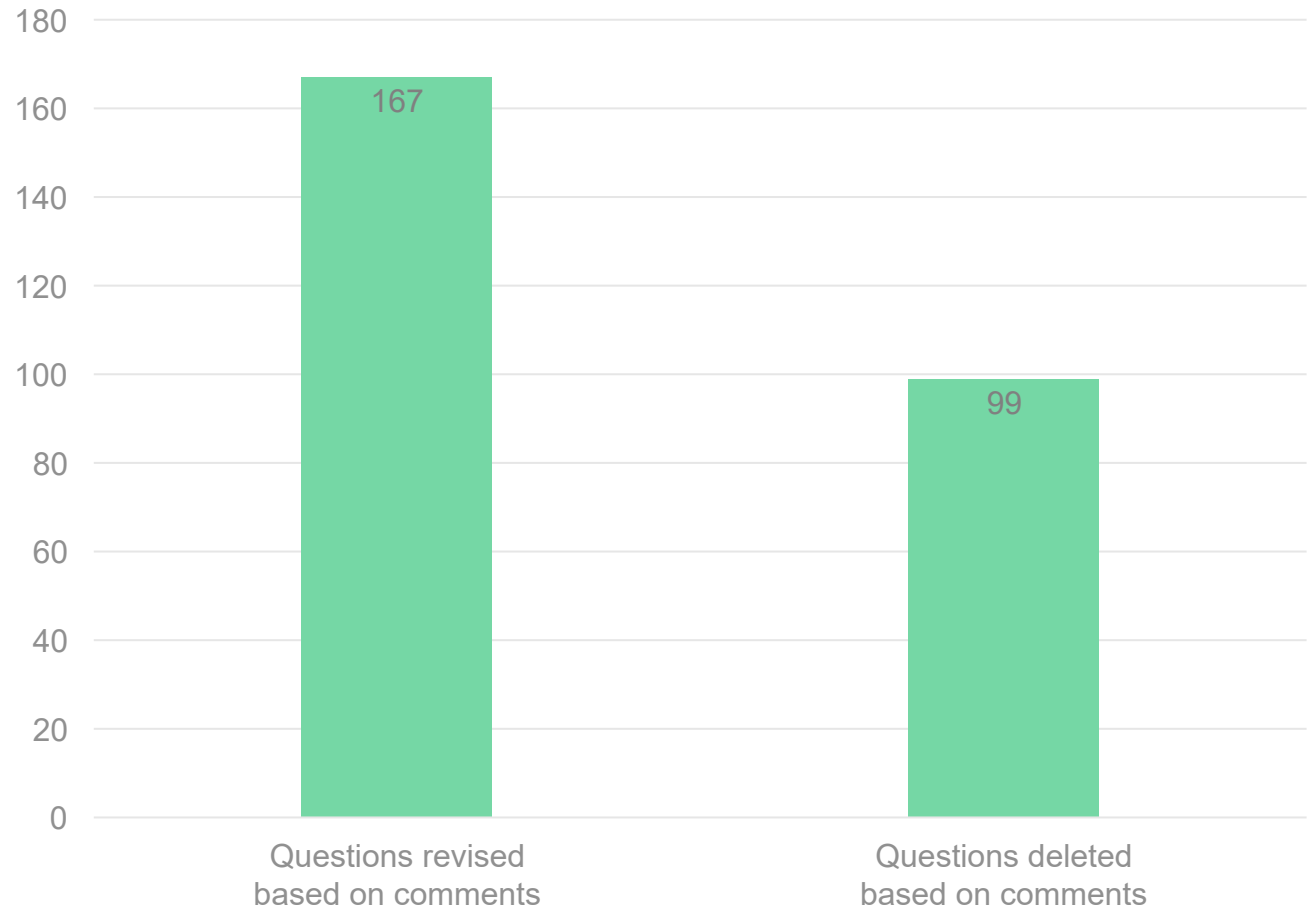
What diplomats told us

Theme	What diplomats said
Inaccurate or Impractical Questions	Answers perceived as incorrect, outdated, or misaligned with current clinical guidelines or real-world anesthesiology practice, including misclassified concepts or unrealistic scenarios.
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Closing the loop on the item pool

Continuous improvement, made visible

- 266 items were revised or retired in a single year because diplomates spoke up.
- Themes point item writers toward the recurring problems, not just individual flaws.
- Findings feed blueprint review, item development training, and content-category planning.



It takes three teams to make one credible report

Assessment Services	Product Management	AI Staff
• Owns item quality and the blueprint • Defines what a theme must mean to be actionable • Supplies ground-truth summaries for validation	• Collects requirements • Oversees schedule • Bridges assessment staff need and technical build	• Designs prompting and model architecture • Runs multi-architecture comparison and tuning • Engineers for speed so iteration is affordable

Working together since January 2026 — a strategic-plan initiative, not a side project.

A trusted partner in lifelong learning

Publishing what we heard — and what we changed — turns a certification requirement into a conversation.

Transparency

The annual MOCA Minute Feedback Report puts themes and actions in front of diplomates, not just staff.

Accountability

Reporting revisions and deletions shows feedback has consequences for the item pool.

Relevance

Recurring themes keep MOCA Minute anchored to real practice, which is what diplomates ask for most.

Capability

A repeatable, validated pipeline positions the Board for a digital-first assessment ecosystem.

